WORKED SOLUTION

STEP I 2017, Question 1

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q1

The key idea

Each of these integrands is secretly of the form $f'(x)/f(x)$ (or, in the last case, $f'(x)/f(x)^2$), once we clear the trigonometric fractions. The derivative of $x\sin x + \cos x$ is unusually tidy, because the two $\sin x$ terms cancel, and the other substitutions are built on the same cancellation. So the plan is: rewrite each integrand in terms of sin and cos (or sec and tan), spot the denominator whose derivative matches the numerator, and substitute.

Solution

Part (i)

With $u = x\sin x + \cos x$, the product rule gives

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sin x + x\cos x - \sin x = x\cos x.$$

Now multiply the top and bottom of the integrand by $\cos x$:

$$\frac{x}{x\tan x + 1} = \frac{x\cos x}{x\sin x + \cos x}.$$

The numerator is exactly $\mathrm{d}u/\mathrm{d}x$, so $x\cos x\,\mathrm{d}x = \mathrm{d}u$ and

$$\int \frac{x}{x\tan x + 1}\,\mathrm{d}x = \int \frac{1}{u}\,\mathrm{d}u = \ln|u| + C = \ln|x\sin x + \cos x| + C.$$

For the second integral we do the same thing with $\sin x$ in place of $\cos x$. Multiplying top and bottom by $\sin x$,

$$\frac{x}{x\cot x - 1} = \frac{x\sin x}{x\cos x - \sin x}.$$

Take $v = x\cos x - \sin x$. Then

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \cos x - x\sin x - \cos x = -x\sin x,$$

so the numerator is $-\mathrm{d}v/\mathrm{d}x$ and

$$\int \frac{x}{x\cot x - 1}\,\mathrm{d}x = -\int \frac{1}{v}\,\mathrm{d}v = -\ln|x\cos x - \sin x| + C.$$

Don't lose the minus sign: it comes from differentiating $\cos x$.

Part (ii)

The first integrand already has a denominator worth trying. Let $w = x\sec^2 x - \tan x$. Using $\frac{\mathrm{d}}{\mathrm{d}x}\sec^2 x = 2\sec^2 x\tan x$,

$$\frac{\mathrm{d}w}{\mathrm{d}x} = \sec^2 x + 2x\sec^2 x\tan x - \sec^2 x = 2x\sec^2 x\tan x.$$

Once again the unwanted terms cancel, and the numerator is $\frac{1}{2}\,\mathrm{d}w/\mathrm{d}x$. Hence

$$\int \frac{x\sec^2 x\,\tan x}{x\sec^2 x - \tan x}\,\mathrm{d}x = \frac{1}{2}\int \frac{1}{w}\,\mathrm{d}w = \tfrac{1}{2}\ln|x\sec^2 x - \tan x| + C.$$

? **Ask yourself** The last integral has no sec or tan in it. How can it be related to $w = x\sec^2 x - \tan x$?

Factor $\cos^2 x$ out of the bracket in the denominator:

$$x - \sin x\cos x = \cos^2 x\left(x\sec^2 x - \frac{\sin x}{\cos x}\right) = \cos^2 x\,(x\sec^2 x - \tan x) = w\cos^2 x.$$

So $(x - \sin x\cos x)^2 = w^2\cos^4 x$, and

$$\frac{x\sin x\cos x}{(x-\sin x\cos x)^2} = \frac{x\sin x\cos x}{w^2\cos^4 x} = \frac{x\sec^2 x\tan x}{w^2},$$

since $\dfrac{\sin x\cos x}{\cos^4 x} = \dfrac{\sin x}{\cos x}\cdot\dfrac{1}{\cos^2 x}$. The numerator is again $\frac{1}{2}\,\mathrm{d}w/\mathrm{d}x$, so with the same substitution

$$\begin{aligned}\int \frac{x\sin x\cos x}{(x-\sin x\cos x)^2}\,\mathrm{d}x &= \frac{1}{2}\int \frac{1}{w^2}\,\mathrm{d}w = -\frac{1}{2w} + C\\ &= -\frac{1}{2(x\sec^2 x - \tan x)} + C = -\frac{\cos^2 x}{2(x-\sin x\cos x)} + C,\end{aligned}$$

where the last step multiplies top and bottom by $\cos^2 x$. If you prefer double angles, this is $-\dfrac{\cos^2 x}{2x - \sin 2x} + C$.

Checks

- Differentiate each answer and make sure you get the integrand back. For the last one, the quotient rule on $-\cos^2 x/\big(2(x-\sin x\cos x)\big)$ uses $\frac{\mathrm{d}}{\mathrm{d}x}(x-\sin x\cos x) = 1 - \cos 2x = 2\sin^2 x$, and the $\sin^2 x\cos^2 x$ terms combine neatly.

- Numerical evaluation: over $[0.3, 1.2]$ the first integral is $0.34953\ldots$, and $\ln|x\sin x + \cos x|$ changes by the same amount; the last integral over the same interval is $25.7364\ldots$, matching $-\cos^2 x/\big(2(x - \sin x\cos x)\big)$. All four answers were confirmed this way by computer.

Common slips

- Substituting without first clearing $\tan x$ or $\cot x$: the numerator only matches $\mathrm{d}u/\mathrm{d}x$ after multiplying through by $\cos x$ or $\sin x$.
- Dropping the minus sign in the second integral of part (i), or the factor $\frac{1}{2}$ in part (ii).
- Reaching for a logarithm in the last integral out of habit: $1/w^2$ integrates to $-1/w$.

Beyond STEP

The trick here is reverse-engineering: choose the substitution so that the derivative of the denominator, after cancellation, is exactly the numerator. Behind this lies the fact that $\int f'/f = \ln|f|$, the logarithmic derivative, which reappears in complex analysis (counting zeros of a function) and in differential equations as the integrating factor. Many STEP integrals are built by differentiating a carefully chosen function and hiding the answer; Siklos, *Advanced Problems in Mathematics*, has plenty more practice in spotting them.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.